TRAFFIC-JAM

Game Design Document

Transportation-first city builder and civil engineering simulation

Design North Star
Factorio-level optimization, except the machine is a living city. The player builds transportation systems, measures flow, tests changes in Planning Mode, and keeps citizens safe and happy while the city grows without a fixed end state.

Working title	Traffic-Jam
Engine	Rust + Bevy
Primary platform	Linux
Secondary target	Steam Deck
Typical city size	10,000-100,000 population
Scale stress target	10,000,000 population

1. Executive Summary

Traffic-Jam is a top-down, vector-map city builder focused on transportation engineering rather than decorative construction. The player acts as the city civil engineer. Roads, intersections, public transit, parking, construction, maintenance, emergency traffic control, and transportation ordinances are the primary tools. Normal buildings are not manually placed; street frontage is zoned and development emerges from accessibility, land value, density limits, services, policy, and market conditions.

The game is designed around measurable optimization. The player should be able to identify a bottleneck, understand why it exists, simulate a proposed fix, compare before-and-after metrics, commit the project, endure construction disruption, and then observe how citizens and traffic adapt over time.

> **Scale Rule**
> Population count, trip demand, active simulation, and rendering are separate systems. A city of 10 million residents must not create 10 million Bevy entities.

2. Design Pillars

Flow is the game

Traffic should come from believable trips and network constraints. A busy road is not a failure if it moves traffic efficiently; delay, queueing, wasted movement, safety, and reliability matter more than raw volume.

The city is alive

Citizens commute, park, change routes, use transit, react to enforcement, relocate, and leave. Businesses, schools, stadiums, industry, farms, and emergencies create time-dependent demand.

Easy to build, deep to engineer

Basic road drawing and AI-assisted systems should be approachable. Advanced players can tune signal phases, synchronized corridors, construction windows, detours, transit schedules, enforcement, and ordinances.

Simulation scale over graphics

Buildings are represented through zoning and density rather than expensive geometry. The engine spends performance budget on transportation simulation, analytics, and large city scale.

3. Core Gameplay Loop

1. Observe traffic, incidents, complaints, reports, and heatmaps.

2. Diagnose the bottleneck at road, intersection, corridor, neighborhood, or transit level.
3. Design a change to roads, signals, transit, parking, enforcement, or policy.
4. Test the design in Planning Mode with temporary simulation state.
5. Compare current and proposed metrics.
6. Approve the project and schedule construction.
7. Operate through closures, incidents, and adaptation.
8. Measure the result and optimize again.

4. Game Modes and Time

4.1 Endless City

Primary v1 mode. There is no mandatory final victory. The city can continue indefinitely while increasingly difficult efficiency targets and achievements provide long-term goals.

4.2 Scenario Mode

Future mode built around existing transportation problems such as a congested downtown, bridge closure, highway dependency, stadium district, or rapid growth challenge.

4.3 Multiplayer

Future project only. The architecture should keep simulation state, rendering, input, and persistence sufficiently separated to support hosted or locally hosted multiplayer later. Multiplayer must not delay the single-player game.

4.4 Time

- Pause / 1x / 2x / 4x / 8x simulation speeds
- 24-hour day/night cycle at useful speeds
- Weekday versus weekend patterns
- Seasons and weather
- Weekly engineering budget
- Monthly city and growth reports
- Monthly ordinance cycle
- Annual mayor election

5. Scale Architecture

The 10 million population goal is a stress target, not the expected average session size. Typical games are expected to be roughly 10,000 to 100,000 residents, but every core system should avoid assumptions that collapse at metropolitan scale.

5.1 Four Simulation Layers

Layer	Purpose	Representation
Statistical population	Represents the full city	Compact household/population aggregates; no render entities
Trip demand	Creates movement pressure	Origin/destination demand buckets, schedules, mode choice, freight/service demand
Active agents	Simulates trips that require detail	Vehicles/pedestrians only where route-level simulation is useful
Rendered entities	What the player can see	Small visual subset driven by camera and zoom; never the source of truth

5.2 Required Scale Techniques

- Simulation level of detail based on relevance, district activity, and camera proximity.
- Spatial partitioning for road segments, intersections, agents, parking, and incidents.
- Fixed-step simulation decoupled from rendering.
- Data-oriented storage for hot simulation data rather than object-heavy per-citizen structures.
- Batch route demand and aggregate flows when individual pathing does not add gameplay value.
- Cache route trees and corridor costs where safe; invalidate incrementally after network changes.
- Use hierarchical routing for large maps: local streets -> district graph -> arterial/highway graph -> local streets.
- Bound per-frame work. Long calculations use job queues spread across ticks or worker threads.
- Multi-core processing for trip generation, aggregation, analytics, and eligible routing work.
- Camera-driven render culling and aggregation; far traffic becomes flow lines or counts.
- Profile memory and CPU continuously on Steam Deck-class constraints.

5.3 Bevy Entity Policy

Non-negotiable
Do not represent every resident as a persistent Bevy ECS entity. ECS entities are appropriate for active vehicles, intersections, visible incidents, player-selected citizens, and other currently relevant simulation objects. Statistical residents remain in compact simulation storage until promoted.

6. Citizens and Driver Behavior

Citizens conceptually have home, job, salary, age, household, vehicle ownership, preferred transport, work schedule, patience, risk tolerance, happiness, commute satisfaction, income, parking preference, and likelihood of moving. At scale, these attributes may be represented by distributions or compact household records rather than a fully instantiated person object.

6.1 Inspectable Citizens

When the player clicks a person, the game can promote that statistical resident into a richer individual narrative with a randomized name and consistent attributes. Only a small number of inspected citizens need long-lived detailed state. Detailed records may be deleted after the citizen moves out or dies.

6.2 Route Adaptation

- Drivers plan around known congestion at trip start.
- If a new incident occurs after departure, only part of the population responds intelligently.
- Baseline behavioral mix: about 10% highly responsive, 50% patient, 25% impatient, and 15% risky; tune during testing.
- Only about 1-2% of normal drivers should routinely break traffic laws.
- Long waits increase the chance of diversion.
- Route memory decays weekly so drivers may retry a formerly bad route.
- Repeated parking difficulty, enforcement, congestion, and incidents affect future behavior.

6.3 Specialized Vehicles

- City vehicles and buses are generally patient.
- Taxis and rideshare are more impatient.
- Semi-trucks prefer main routes and avoid incompatible underpasses, tight streets, and restricted geometry.
- Emergency vehicles may exceed the speed limit, receive priority, make traffic pull over, and can still cause crashes.

7. Roads, Addresses, Zoning, and Land

Roads define the city. Building a road creates addressable street frontage. Zoning is assigned to the land touching a street segment between intersections rather than painting every individual parcel tile.

- Core zones: Residential, Commercial, Office, Industrial, Farm.
- Player chooses zone type and optional maximum density. The market determines actual density.
- Unzoned land is still classified as park, field, city-owned/open land, or similar and affects value.
- Special service zones include police, fire, hospital, school, stadium, and similar destinations.

- Mixed-use combinations can emerge in dense areas: residential + commercial, residential + office, commercial + office, and three-way mixed use.
- Neighborhood identities emerge and can later receive neighborhood-level ordinances.

7.1 Land Value Inputs

- Commute time
- Congestion
- Transit access
- Noise
- Crime
- Schools
- Parks
- Hospitals
- Fire coverage
- Pollution
- Nearby industry
- Highway access
- Walkability
- Parking availability
- Nearby jobs
- Special attractions
- Traffic accidents
- Property taxes

8. Road and Intersection Engineering

- Hybrid road editor: draw first, then configure engineering details.
- Road classes include local streets, collectors, arterials, boulevards, rural roads, industrial/freight roads, highways, and limited-access freeways.
- Roads support snapping, curves, parallel alignment, 45/90-degree alignment, bridges, tunnels, and layered elevation.
- Terrain elevation is intentionally simple; bridges and tunnels matter more than terrain sculpting.
- Crossing roads create intersections unless designated as grade-separated or disconnected.

8.1 Intersection Controls

- Through and turn lanes
- Protected turns
- Yield movements
- Turn restrictions
- Crosswalks and pedestrian signals

- 2-way, 3-way, and 4-way stops
- Yield and uncontrolled intersections
- Roundabouts with lane, entry, exit, crossing, truck apron, and speed configuration

8.2 Traffic Signals

- Cycle length
- Individual phases
- Green/yellow/all-red timing
- Protected turns
- Leading pedestrian intervals
- Detector actuation
- Transit priority
- Emergency preemption
- Time-of-day schedules
- Event plans
- Corridor synchronization and green waves

9. Parking, Walking, Cycling, and Transit

- Parking lots, garages, street parking, paid parking, residential parking, park-and-ride, and illegal parking.
- When parking is full, drivers search outward for the next available space, creating measurable search traffic.
- Sidewalks are configurable road features and affect walkability, safety, accessibility, and land value.
- Cycling supports bike lanes, protected lanes, bike paths, shared roads, parking, and later bike-share systems.
- Transit includes buses, subway, passenger rail, and freight rail. Transit can be manual, AI-assisted, or fully automated.
- Underground view fades surface roads to low opacity while showing subway and tunnel infrastructure prominently.

10. Vehicles, Fuel, and External Connections

- Personal cars, motorcycles, taxis, rideshare, buses, school buses, semis, box trucks, delivery vans, garbage trucks, farm equipment, police, fire, ambulance, tow, construction, and utility vehicles.
- Households can own specific vehicles based on income, density, parking, transit, household size, policy, and preference.
- Fuel/charge state can create trips to gas stations or EV chargers. EV adoption is driven primarily by policy and ordinances.

- External highways, roads, and rail are sources/sinks only. The world outside city limits is not simulated.

11. Economy and Development

Traffic-Jam intentionally avoids a detailed commodity supply-chain simulation. Industry creates jobs and freight traffic; commercial creates customer activity and delivery demand; offices create service-sector employment and revenue. Transportation quality directly changes business performance.

- Businesses track enough state to react to employee access, customers, freight reliability, parking, transit, and cost.
- Poor access can shrink, relocate, or close businesses.
- Farms use large areas, generate seasonal traffic, and can be converted to development as land values rise.
- Citizens initially prefer jobs within roughly five miles, then trade commute against salary and may later move closer to better jobs.

12. Safety, Police, Incidents, and Disasters

- Crash risk comes from speed, driver behavior, lane changes, intersection design, signal violations, weather, congestion, construction, road condition, and emergency vehicles.
- Crash severity ranges from minor incidents through disabled vehicles, injuries, multi-car crashes, and fatal crashes.
- Police, fire, ambulance, tow, and road crews physically travel through the network.
- Police can perform traffic and parking enforcement. More enforcement improves safety and revenue but can reduce citizen happiness.
- Drivers remember recent enforcement locations and may slow down there on later trips.
- Traffic Control Mode lets the player close roads, create detours, modify signal timing, restrict trucks, redirect transit, or assign police to intersections.

12.1 Event Catalog

- Vehicle breakdowns
- Police chases
- Major crashes
- Truck spills
- Train/subway failures
- Signal failure
- Power outage
- Flooding
- Snow and ice
- Heavy rain and severe storms
- Sporting events and concerts

- Parades and protests
- Motorcades
- Sinkholes
- Bridge inspections and closures
- Emergency repairs

13. Construction and Maintenance

Approved infrastructure takes time to build, modify, or demolish. Basic construction targets roughly one day of work per lane before complexity modifiers, subject to balancing.

- Limited city crews and contractor options create project capacity constraints.
- Contractor tiers trade cost, speed, reliability, and availability.
- Beginner scheduling can be automatic; advanced players can control start dates, work windows, lane closures, phases, detours, temporary signals, temporary limits, and police support.
- Projects can suffer weather delays, contractor delays, material shortages, work-zone crashes, complaints, overruns, failed inspections, and emergency crew reallocation.
- Road condition uses a simple Excellent -> Good -> Worn -> Poor -> Critical model.
- Higher sustained maintenance funding slows deterioration and reduces major repairs.
- Repair quality choices: Cheap, Balanced, Premium.

14. Planning Mode

Signature Feature
Planning Mode creates a temporary simulation branch where proposed roads, lane changes, signals, and other network edits behave as if complete. The player can run the simulation, compare metrics, change the plan, and then either discard it or approve construction.

- Unlimited/forgiving undo while planning.
- Planning state does not replace the real save state.
- Plan Comparison Report shows commute, delay, intersection efficiency, crash risk, transit delay, parking behavior, construction cost, maintenance impact, land-value impact, and expected construction duration.
- After construction is complete, removal requires real demolition time and cost.

15. Weather, Calendar, and Scheduled Operations

- Rain, heavy rain, snow, ice, fog, heat, flooding, and severe storms affect traffic and operations.
- Snow management includes plow/salt priorities, parking bans, emergency routes, school routes, and transit routes.
- Weekly, monthly, and annual calendar views show events, closures, construction, inspections, weather, school schedules, and transit changes.

- The calendar is also the control surface for scheduled signal plans, detours, transit timings, event plans, and automations.

16. Budget, Mayor, Ordinances, and Politics

The player receives a weekly engineering budget representing a share of city resources. The player decides how this budget is split between maintenance, new roads, transit, signals, bridges, and emergency work.

- Mayor personalities can prioritize growth, environment, business, transit, low taxes, safety, or suburban development.
- Annual mayor elections can change strategic priorities.
- The player does not directly set city tax rates; tax and broader policy changes are proposed through ordinances.
- Ordinance support depends on citizens, council, city conditions, and the player's engineering reputation.
- Examples include parking fees, congestion pricing, truck restrictions, HOV/bus lanes, speed limits, school zones, EV incentives, residential permits, parking minimums, cycling incentives, and work-from-home incentives.
- Neighborhood-specific ordinances are supported.

17. Analytics and Optimization

Analytics are a primary gameplay system, not a decorative dashboard. The player should be able to identify the worst-performing transportation elements and iterate like a factory optimization game.

- City Engineering Score plus category scores for traffic, safety, happiness, economy, transit, infrastructure, environmental impact, and growth.
- Per-road and per-intersection efficiency scores.
- Vehicles/hour, average delay, maximum queue, capacity utilization, crash rate, turn movement, truck share, transit delay, emergency delay, and peak periods.
- Wasted movement: parking search, empty rideshare repositioning, truck detours, buses in traffic, poor transfers, routing inefficiency, and emergency delay.
- Historical comparisons by day, week, and month.

17.1 Heatmaps

- Congestion
- Average speed
- Crash risk
- Actual crashes
- Driver frustration
- Noise

- Pollution
- Land value
- Parking demand
- Transit demand and coverage
- Emergency response
- Freight
- Pedestrians
- Road condition
- Construction impact
- Density
- Jobs
- Population

18. Social Feed, Alerts, and Reports

- Alert severity: Info, Advisory, Warning, Critical.
- Only severe events should be candidates for automatic pause, subject to settings.
- Social media posts are derived from real simulation events. Tone can be humorous or exaggerated, but information is never fabricated.
- Clickable posts can navigate to the relevant street, intersection, route, district, or incident.
- Monthly reports summarize population, score, commute, delay, crashes, tickets, happiness, transit, road condition, construction, emergency response, land value, business activity, revenue, budget, ordinances, and neighborhood issues.

19. Visual and Audio Direction

- Top-down vector-map presentation inspired by the clarity of modern navigation maps, not satellite imagery.
- No detailed building models. Parcel/zoning color communicates land use; greater saturation communicates greater density.
- Cars are simple dots; trucks and buses use rectangular marks; taxis/rideshare use yellow; emergency/service vehicles use distinct simple symbols or colors.
- At distant zoom, individual vehicles disappear into aggregate traffic flow.
- Maximum zoom-out emphasizes major roads, transit, districts, congestion, value, population, construction, and incidents.
- Audio is restrained: traffic ambience, rail, sirens, construction, weather, UI alerts, and light ambient music.

20. Difficulty, Failure, and AI Automation

- Easy: relaxed building, larger margins, more patient drivers, slower deterioration, fewer severe incidents.

- Normal: balanced.
- Hard: tighter budgets, more operational incidents, faster pressure, less forgiving traffic.
- Expert: maximum legitimate simulation pressure without AI cheating.
- Failure may result from financial collapse, population collapse, critical happiness, political removal, unsafe infrastructure, or fiscal emergency.
- AI automation should be competent and conservative. Skilled human players can outperform it through better design rather than receiving arbitrary manual-control bonuses.

21. Technology and Progression

Core technology does not advance through historical eras. EV adoption changes through policy and ordinances. Autonomous vehicles can have distinct behavior, but connected smart-intersection systems are a later project. Meta-progression, unlock trees, and broad technology progression are stretch goals rather than v1 requirements.

22. Maps, Saves, and Modding

- Procedural seeded maps with rivers/lakes, external connections, small existing town, farmland, industry, parks, and open land.
- Starting presets can include rural town, suburban sprawl, dense core, industrial city, transit-heavy city, and highway-dependent city.
- City boundaries expand automatically as development approaches the edge.
- Manual saves, autosaves, multiple save slots, named checkpoints, and planning snapshots.
- Human-readable RON/JSON for metadata/configuration where practical; compressed binary state for large simulations.
- Future mod architecture should favor safe data-driven extension points for roads, vehicles, ordinances, events, scenarios, maps, overlays, and simulation parameters.

23. Technical Architecture

23.1 Runtime Domains

- Network graph: roads, lanes, intersections, turn movements, controls, restrictions, geometry layers.
- Population simulation: household aggregates, employment aggregates, migration, happiness, ownership, policy response.
- Trip demand: time-of-day origin/destination demand, mode choice, special events, external sources/sinks.
- Routing: hierarchical graph, cached costs, restrictions, incident overlays, vehicle-class constraints.
- Active mobility: lane movement, queues, intersection entry, parking search, transit vehicles, emergency priority.

- City economy: development pressure, business viability, land value, city revenue, engineering budget.
- Operations: construction, maintenance, incidents, weather, police, emergency response, ordinances.
- Analytics: rolling counters, histograms, heatmap tiles, historical summaries, plan comparisons.
- Presentation: map rendering, overlays, UI, social feed, alerts, calendar, reports.

23.2 Threading and Tick Strategy

- Use a fixed simulation tick separate from render frames.
- Different systems may update at different cadences: lane movement frequently; trip demand less often; land value and economic development much less often.
- Partition the city into spatial regions so active simulation work can be parallelized safely.
- Keep writes localized where possible and use staged command buffers for cross-region changes.
- Do not recompute global analytics every tick; use incremental counters and scheduled rollups.
- At high game speed, prefer coarser simulation representation rather than simply executing eight times as many expensive detailed ticks.

23.3 Routing Strategy

- Vehicle-class-aware restrictions for trucks, buses, emergency vehicles, and private cars.
- Hierarchical route graph separates local navigation from high-capacity corridor navigation.
- Cache stable route subtrees and invalidate on topology, closure, or meaningful cost change.
- Use approximate/aggregated assignment for distant flows; instantiate detailed routes near active regions or when the player inspects a trip.
- Planning Mode should use copy-on-write or overlay network state rather than duplicating the entire city.

23.4 Steam Deck Optimization

- Graphics quality presets must independently control map effects, antialiasing, labels, particles, and visible vehicle density.
- Simulation presets control active-agent radius, routing budget, aggregation thresholds, and analytics cadence without changing core rules more than necessary.
- Provide adaptive simulation LOD based on measured frame/simulation budget.
- Target stable interaction first; visual density should degrade before core city state does.

24. MVP and Development Order

24.1 MVP Required Systems

- Procedural starting town
- Road drawing/editing

- Intersections, traffic lights, stop signs
- Residential/commercial/office/industrial/farm zoning
- Basic commuting citizens
- Cars, buses, trucks, emergency vehicles
- Parking
- Congestion
- Accidents and breakdowns
- Basic emergency response
- Weekly engineering budget
- Basic road maintenance
- Planning Mode
- Traffic heatmap
- Road/intersection analytics
- Basic social feed
- Save/load

24.2 Systems Allowed to Be Simplified Initially

- Rail/subway depth
- Contractor complexity
- Weather detail
- Advanced ordinances
- Mayor/election depth
- Full pedestrian/cycling simulation
- Police enforcement depth
- Neighborhood identity
- Advanced mixed-use redevelopment

24.3 Milestone Sequence

Milestone	Goal	Exit condition
1. Tiny complete city	Prove the loop	Road -> zone -> residents/jobs -> trips -> congestion -> inspect -> redesign -> measurable improvement
2. Road editor	Make engineering pleasant	Fast drawing, lane/intersection edits, planning overlays, useful undo
3. Traffic simulation	Deepen behavior	Routing, lane flow, parking, adaptation, incidents, vehicle

		classes
4. Analytics	Make optimization legible	Heatmaps, historical metrics, bottleneck ranking, plan comparisons

25. First Vertical Slice

The first serious prototype should be intentionally ugly and small. Target roughly 500-2,000 population. It is successful when the following loop works reliably:

9. Generate or open a simple map.
10. Draw two roads and create an intersection.
11. Zone homes and jobs.
12. Residents appear and commute.
13. A traffic signal controls the intersection.
14. Insufficient capacity creates measurable congestion.
15. The player enters Planning Mode and changes the intersection or road.
16. The proposed simulation shows improvement or regression.
17. The player approves the change.
18. Analytics prove the result after construction.

Do Not Start With
Beautiful UI, subway, elections, deep weather, detailed construction, multiplayer, advanced police systems, or a 10-million-person simulation. Prove the complete transportation loop first.

26. Success Criterion

Traffic-Jam succeeds if a player can spend an hour studying one area of the city and think, "I know I can make this intersection 5% better," then redesign it, test it, deploy it, and watch the surrounding city react.

The ideal end state feels like watching a highly optimized factory: cars flow, signals synchronize, buses arrive on time, trucks stay on freight corridors, emergency vehicles get through, parking is available, citizens arrive on time, accidents are rare, and construction is planned. Then growth creates the next bottleneck.

27. One-Sentence Pitch

Pitch
Traffic-Jam is a Rust-based city transportation simulator where you engineer an endlessly growing city, optimize every movement from individual traffic lights to metropolitan transit

networks, and pursue Factorio-level efficiency without making your citizens miserable.